# Haebeom **Jung**

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## Education

**Seoul National University** — *Seoul, South Korea*

Ph.D. Student in Artificial Intelligence — *Sep. 2023 - Present*

- **Current GPA**: 4.04/4.30
- A member of the SNU VGI Lab (Advisor: Prof. Jaesik Park)
- Received full scholarship for 5 semesters.

**KwangWoon University** — *Seoul, South Korea*

B.S. in Division of Robotics — *Mar. 2016 - Feb. 2023*

- **GPA**: 3.81/4.50, **Major GPA**: 3.93/4.50
- Received full scholarship for 6 semesters.

## Publications

***Beyond the Current Scene: Event-Referential Grasping with Active View Selection*** — *arXiv 2026*

Hyunjoon Lee*, **Haebeom Jung***, Eunsung Cha, Daeun Lee, Yu-Chiang Frank Wang, Jaesung Choe, and Jaesik Park — *Sep. 2026*

(* equal contribution)

**Distractor-Aware COLMAP-Free 3D Gaussian Splatting** — *Under Review*

Namtae Kim, **Haebeom Jung**, and Jaesik Park

**Articulated Object Reconstruction from Rest-State Observation** — *ECCV 2026*

Daeun Lee, Jaeah Lee, Woosung Kim, **Haebeom Jung**, and Jaesik Park — *Sep. 2026*

***Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting*** — *RA-L 2026*

**Haebeom Jung**, Namtae Kim, Jungwoo Kim, and Jaesik Park — *Apr. 2026*

***Metropolis-Hastings Sampling for 3D Gaussian Reconstruction*** — *NeurIPS 2025*

Hyunjin Kim, **Haebeom Jung**, and Jaesik Park — *Dec. 2025*

## Professional Experience

**Kakao Mobility** — *Pangyo, South Korea*

Software · Robotics Engineer (Full-time) — *Dec. 2021 - Apr. 2022*

- Developed a data collection system for autonomous vehicles.
- Researched visual SLAM and localization for real-world driving environments.

**Stryx** — *Seoul, South Korea*

Software · Robotics Engineer (Compulsory Military Service) — *Dec. 2019 - Dec. 2021*

- Led software development for the hardware business unit.
- Developed a sensor data collection system for a mobile mapping platform.
- Researched and developed autonomous navigation for unmanned ground vehicles, including SLAM, localization, sensor fusion, and path planning.

**ROBOTIS** — *Seoul, South Korea*

Robotics Engineer (Intern) — *Jan. 2019 - Feb. 2019*

- Validated ROBOTIS OpenManipulator prior to its open-source release.
- Developed and published an example application for ROBOTIS OpenManipulator SARA.

## Skills

**Programming** Python, C/C++, Java
**Robotics & Tools** PyTorch, ROS/ROS 2, Isaac Sim, Qt
**Languages** Korean (native), English (fluent)

## Open Source Contributions

### Selected Robotics Projects

- Co-author of **BeyondCSe**, the official implementation of Event-Referential Grasping with Active View Selection (arXiv 2026).
- Author of **TLC-Calib**, the official implementation of Targetless LiDAR-Camera Calibration with Neural Gaussian Splatting (RA-L 2026).
- Maintainer of **ORB-SLAM3-ROS2**, a ROS 2 integration for the ORB-SLAM3 library.
- Contributor to **FAST-LIO**, fixing Velodyne LiDAR integration issues.
- Contributor to **LIO-SAM**, adding a ROS service for map saving.

## Patents

***Device and Method for Targetless LiDAR-Camera Calibration*** — *Seoul National Univ., South Korea*
Haebeom Jung, Namtae Kim, and Jaesik Park — *May. 2025*
*Application No. 10-2025-0067792*

***Method of estimating the location of a moving object using vector map*** — *Kakao Mobility, South Korea*
Yesung Hwang, Haebeom Jung, Jeongmin Kang, Ilsuk Park, Seungkwan Song, and Seunghwan Hong — *Jan. 2024*
*Registration No. 10-2624-6440000*

## Awards

### Robotics Competitions

International

| | | |
|---|---|---|
| 2018 | **3rd Place**, ROBOCUP CANADA, KidSize Humanoid Technical Challenge **[Video]** | *Montreal, Canada* |
| 2017 | **4th Place**, ROBOCUP IRAN, KidSize Humanoid Soccer **[Video]** | *Tehran, Iran* |

Domestic

| | | |
|---|---|---|
| 2018 | **3rd Place**, R-BIZ Challenge, Turtlebot3 Autorace **[Video]** | *Ilsan, South Korea* |
| 2018 | **2nd Place**, IRC, Intelligence Humanoid Robot Sports | *Ilsan, South Korea* |
| 2017 | **Winner**, IRC, Intelligence Humanoid Robot Sports **[Video]** | *Ilsan, South Korea* |
| 2017 | **Winner**, ROBOCUP KOREA, Intelligence Humanoid Soccer | *Daegu, South Korea* |
| 2016 | **3rd Award**, KIRO, Future-Robot Idea Presentation | *Pohang, South Korea* |

### Honors

| | | |
|---|---|---|
| 2019 | **Outstanding Mentor**, 2019 Samsung Electronics Junior Software Creation Contest | *Samsung Electronics* |
| 2017 | **Achievement Award**, Spread the name of KwangWoon University throughout the year | *KwangWoon Univ.* |

### Scholarships

| | | |
|---|---|---|
| 2026 | **Full scholarship**, Jeongheon Foundation Scholar | *Jeongheon Foundation* |
| 2023-2025 | **Full scholarship**, AI Fellowship | *Seoul National Univ.* |
| 2017-2019 | **Full scholarship**, Robotics Competition Team | *KwangWoon Univ.* |

## Fellowships and Research Funding

**Photo-Realistic Rendering for Surrounding Dynamic Mobile Scenes** — *Ministry of Education, South Korea*
Master's Research Fellowship Program — *Jul. 2024 - Jun. 2025*

**Artificial Intelligence Graduate School Program** — *Ministry of Science and ICT, South Korea*
Seoul National University, IPAI — *Sep. 2023 - Jul. 2025*

# Research Service & Robotics Activities

### Invited Talks

- NAVER LABS, Apr. 2025

### Reviewer

- CVPR (2026)
- ECCV (2026)
- NeurIPS (2026)
- IEEE TIP (2025)

### Kakao Mobility NEMO Conference

*Seoul, South Korea*

Autonomous driving robot docent

*Feb. 2022*

- Explained and operated an autonomous robot that travels inside the conference hall. **[Video]**

### Samsung Electronics Junior Software Creation Contest

*Suwon, South Korea*

Mentor

*Jul. 2019 - Sep. 2019*

- Serviced the app, which provided a system for managing kindergarten students going to and from school and attendance using beacons.
- The high school team I coached won the grand prize among 10,000 teams.

### RO:BIT (Robot Game Sports Team)

*Seoul, South Korea*

Member & Team Leader at 2018

*Jul. 2016 - Dec. 2018*

- As a member of the humanoid team for 3 years, I participated in many external activities and competitions related to robots.
- I mainly researched and developed a robot algorithm called humanoid intelligence in games such as soccer, basketball and athletics, etc.
- In 2018, I led the team as the leader of the humanoid team and won awards in international competitions.